

DATA ANALYTICS WITH TABLEAU

Project title : Toycraft tales: tableau's vision into toy manufacturer data.

UNIT-I : Introduction to BI & Tableau

Understand the concept of Business Intelligence - Explain the process of data integration, processing, and presentation - Familiarize with ETL (Extract, Transform, Load) architecture- Gain an introduction to data analytics and its types - Understand descriptive, diagnostic, predictive, and prescriptive analytics-Explore the applications of analytics in business.

Introduction to Tableau :

Introduction to tableau-Overview & Features–Connecting Tableau to Data Sources -Working with Flat files-Connecting spreadsheets

Data Extraction:

Understand the fundamentals of databases and their role in data management - Learn how to create databases and tables for data storage - Develop skills in performing CRUD (Create, Read, Update, Delete) operations on database tables - Gain a basic understanding of SQL operations for data extraction and manipulation

Architecture of Tableau :

Gain knowledge of the architecture of Tableau and how it functions - Explore the interface of Tableau, including layout, toolbars, data pane, and analytics pane - Understand different Tableau field types and their usage - Learn how to save and publish data sources in Tableau - Differentiate between live and extract

connections in Tableau - Explore various file types and ways to share and export work done in Tableau.

Objective: Foundations & Data Acquisition.

Understanding the toy manufacturing landscape and preparing the raw data.

Industry Overview: Identifying key players in toy manufacturing and the specific business challenges (seasonality, safety regulations, and shipping costs).

Data Sourcing: Introduction to the datasets (often CSV/Excel) containing sales records, factory production logs, and inventory levels.

Defining KPIs: Determining what "success" looks like—Metrics such as Gross Margin, Defect Rate, and Inventory Turnover.

UNIT-II : Data Visualization & Data Blending

Understand different chart types for data visualization, including histograms, box plots, pie charts, bar charts, line charts, bubble charts, bullet charts, scatter plots, tree maps, heat maps, maps, text tables, and highlighted tables - Learn how to create effective visualizations using Tableau

Working with Metadata & Data Blending:

Connect to different data sources in Tableau - Understand Tableau data types - Learn how to connect to Excel, cubes, and PDFs in Tableau - Manage metadata and extracts in Tableau.- Master joining techniques such as left, right, inner, and outer joins, as well as union operations - Explore data blending and data preparation techniques in Tableau.

Objective: Data Preprocessing (The Cleanup)

Transforming messy manufacturing logs into "Tableau-ready" formats.

Data Cleaning: Handling null values in factory logs and correcting inconsistent toy category names using Tableau Prep.

Data Integration: Using Joins and Unions to connect different tables, such as linking "Product IDs" to "Regional Sales" data.

Schema Design: Understanding how to structure data for time-series analysis (crucial for toy sales which peak in Q4).

UNIT-III : Advanced-Data Manipulations & Filters

Learn advanced techniques for marking and highlighting data points - Understand how to group data and create sets in Tableau - Explore bins, hierarchies, and sorting options in Tableau - Utilize the formatting pane for menu, font, alignment, and other settings - Edit axes and annotations in Tableau

Working with Filters, Organizing Data & Visual Analytics:

Gain proficiency in working with filters in Tableau - Learn different types of filters and how to add or remove them - Explore filtering techniques for continuous dates, dimensions, and measures - Understand the order of operations in filtering data - Learn techniques for organizing data and conducting visual analytics in Tableau.

Working With Mapping, Calculations, and Expressions:

Gain proficiency in working with filters in Tableau - Learn different types of filters and how to add or remove them - Explore filtering techniques for continuous dates, dimensions, and measures - Understand the order of operations in filtering data - Learn techniques for organizing data and conducting visual analytics in Tableau

Working With Mapping, Calculations, and Expressions:

Work with coordinate points and plot longitude and latitude data on maps in Tableau -Edit unrecognized locations and work with background images in Tableau - Understand map visualization and custom territories in Tableau - Learn calculation syntax and functions in Tableau. - Create calculated fields and utilize quick table calculations

Working with Parameters :

Gain knowledge of creating parameters in Tableau - Understand how to use parameters in calculations - Learn how to incorporate parameters with filters in Tableau.

Objective: Building the "Vision" through interactive Tableau worksheets.

Geographic Mapping: Creating symbol maps to visualize where toys are manufactured vs. where they are sold.

Trend Analysis: Using Line Charts to track manufacturing costs and sales spikes across different years.

Performance Benchmarking: Building Bar Charts and Heat Maps to compare the performance of different toy lines (e.g., Action Figures vs. Educational Toys).

Calculated Fields: Writing formulas in Tableau to calculate "Net Profit" or "Days to Ship."

Unit 4: Storytelling & Business Insights

Objective: Communicating findings to stakeholders through Tableau Stories.

Dashboard Interactivity: Implementing Filters and Actions so users can click a specific region to see its manufacturing defects.

The Narrative: Organizing dashboards into a Tableau Storyboard that leads the viewer from a "Global View" down to "Factory-Specific Issues."

Actionable Insights: Formulating recommendations based on the data, such as "Reduce production of X toy in Q1 to save on storage costs."

Unit 5: Deployment & Portfolio.

Objective: Finalizing the project for professional review.

Optimization: Improving dashboard load times and ensuring the UI is mobile-friendly.

Publishing: Uploading the final workbook to Tableau Public to showcase skills to recruiters.

Documentation: Drafting the project report detailing the insights discovered during the analysis.

Project Flow

To accomplish this, we have to complete all the activities listed below,

Data Collection & Extraction from Database

- o Collect the dataset,
- o Storing Data in DB
- o Perform SQL Operations
- o Connect DB with Tableau
- o Data Preparation
- o Prepare the Data for Visualization

Data Visualizations

- o Number of Unique Visualizations

Dashboard

- o Responsive and Design of Dashboard

Story

- o No of Scenes of Story

Performance Testing

- o Amount of Data Rendered to DB ‘
- o Utilization of Data Filters

- o No of Calculation Fields
- o No of Visualizations/ Graphs

Web Integration

- o Dashboard and Story embed with UI With Flask

Project Demonstration & Documentation

- o Record explanation Video for project end to end solution
 - Project Documentation-Step by step project development procedure

No of Unique Visualizations

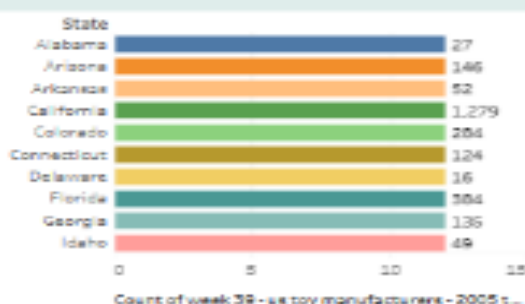
The number of unique visualizations that can be created with a given dataset. Some common types of visualizations that can be used to analyze the performance and efficiency of ToyCraft Tales: Tableau's Vision into Toy Manufacturer Data include bar charts, line charts, heat maps, scatter plots, pie charts, Maps, etc. These visualizations can be used to compare performance, track changes over time, show distribution, and relationships between variables, breakdown of revenue and customer demographics, workload, resource allocation, and location of hotels.

Responsive and Design of Dashboard

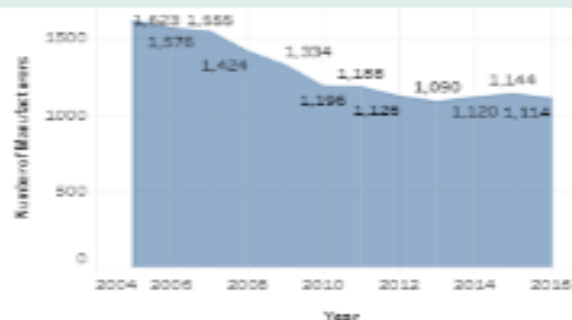
The responsiveness and design of a dashboard for analyzing the performance ToyCraft Tales: Tableau's Vision into Toy Manufacturer Data is crucial to ensure that the information is easily understandable and actionable. Key considerations for designing a responsive and effective dashboard include user-centered design, clear and concise information, interactivity, data-driven approach, accessibility, customization, and security. The goal is to create a dashboard that is user-friendly, interactive, and data-driven, providing actionable insights to improve the performance ToyCraft Tales: Tableau's Vision into Toy Manufacturer Data. Once you have created views on different sheets in Tableau, you can pull them into a dashboard.

ToyCraft Tales Tableau's Vision into Toy Manufacturer Data

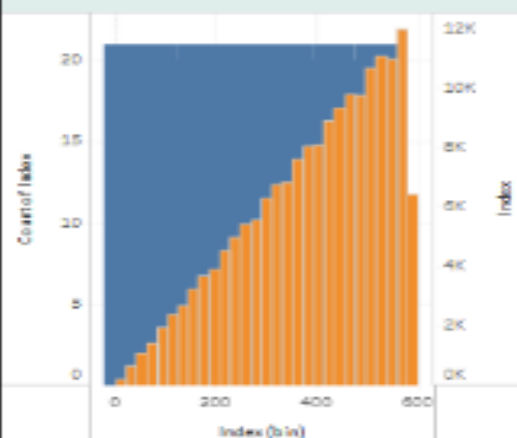
Top 10 state Toy Manufacturers in US States



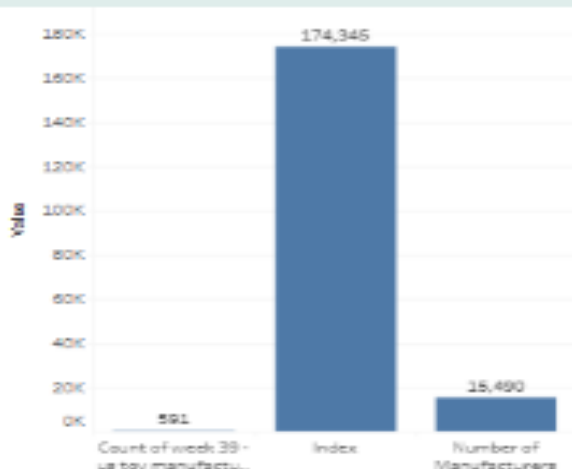
Analysis on number of Manufacturers by year



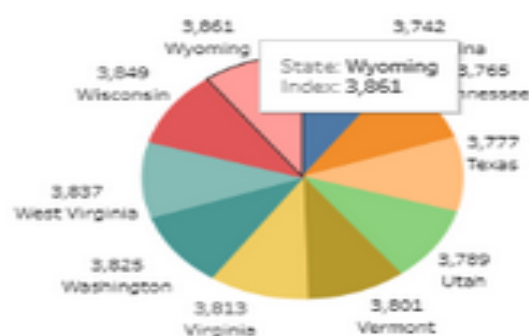
Analysis on Toy Manufacturers by Index



Analysis on Toy Manufacturers



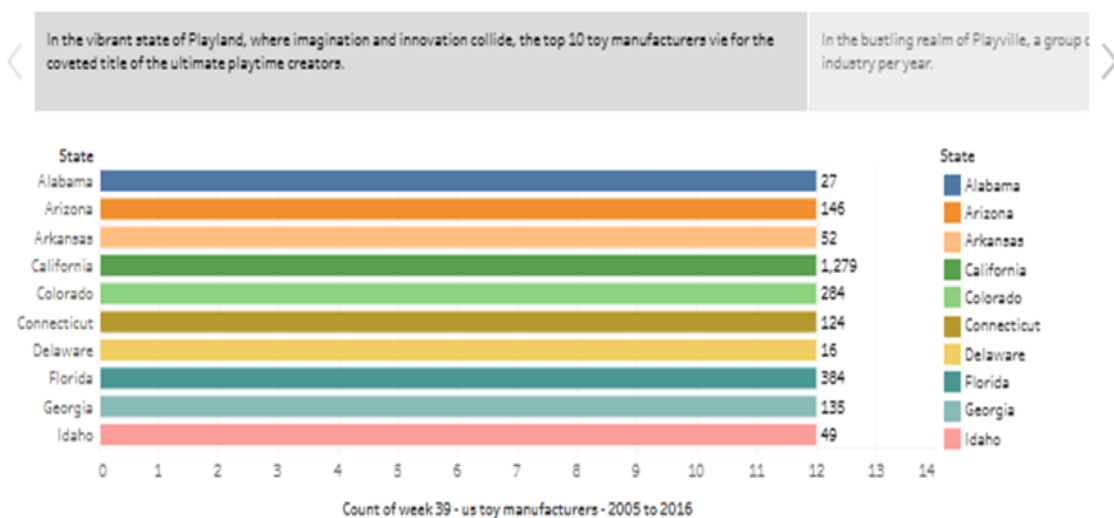
Analysis on Toy Manufacturers in US States by Index



No of Scenes of Story

The number of scenes in a storyboard for a data visualization analysis of ToyCraft Tales: Tableau's Vision into Toy Manufacturer Data will depend on the complexity of the analysis and the specific insights that are trying to be conveyed. A storyboard is a visual representation of the data analysis process and it breaks down the analysis into a series of steps or scenes.

ToyCraft Tales Tableau's Vision into Toy Manufacturer Data story



Amount of Data Rendered to DB

The amount of data that is rendered to a database depends on the size of the dataset and the capacity of the database to store and retrieve data.

Open the MySQL Workbench, go to the database then click to expand the tables, select the table, and click on the (i) button to get the information related to the table such as column count, table rows etc.

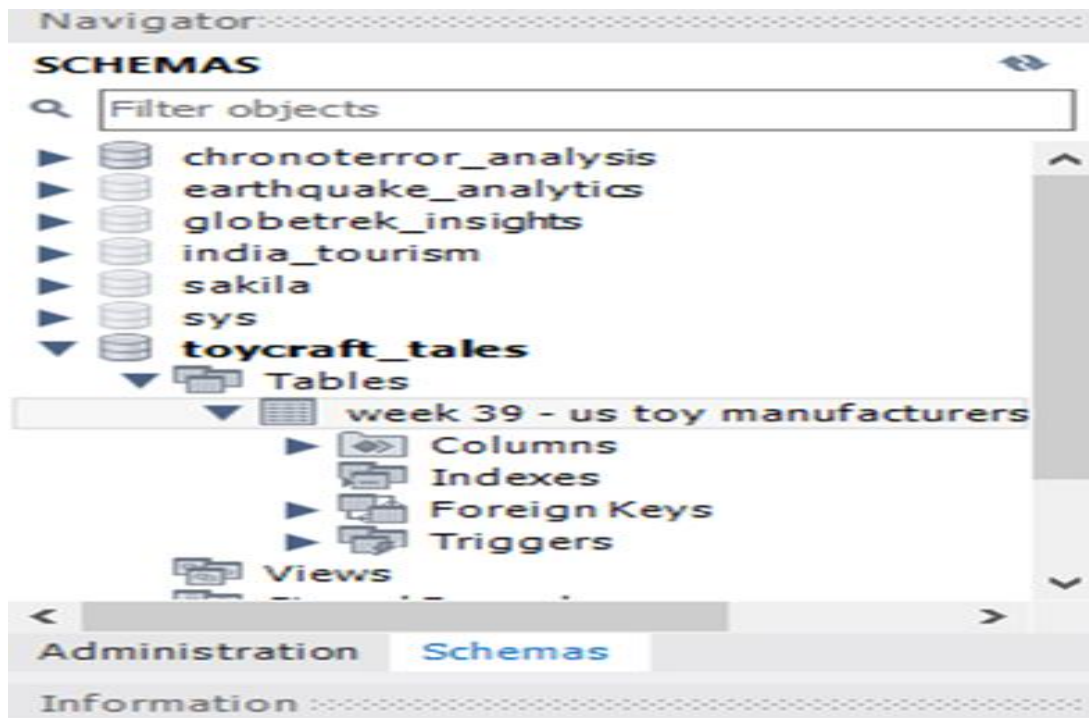
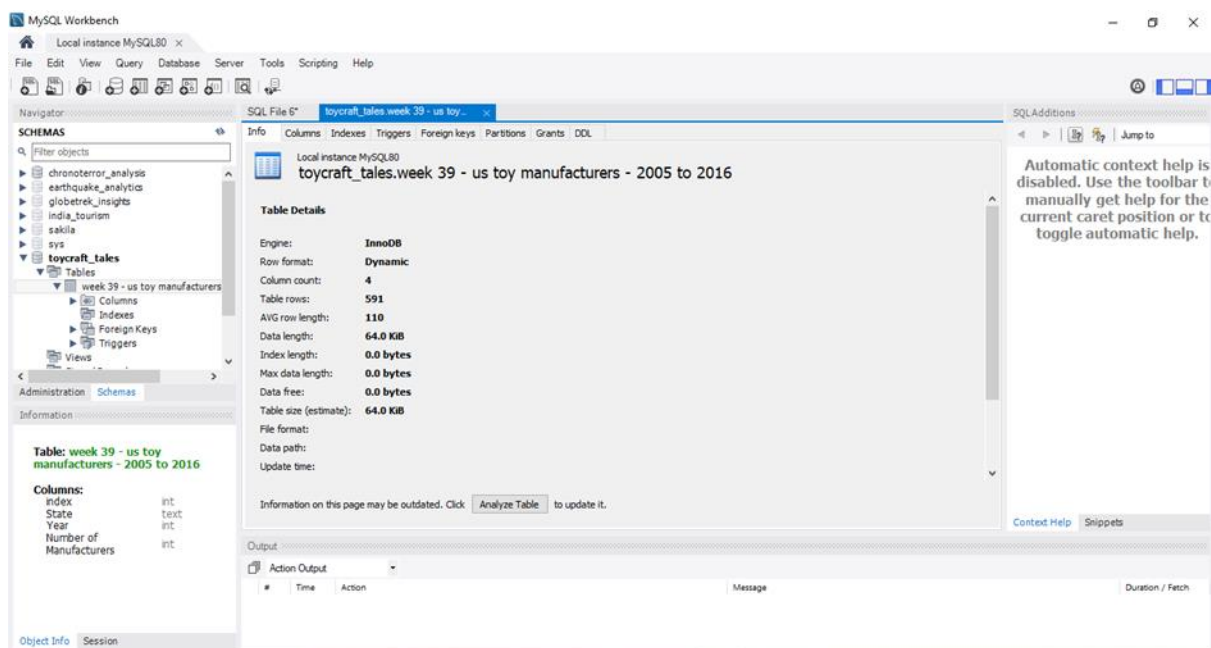


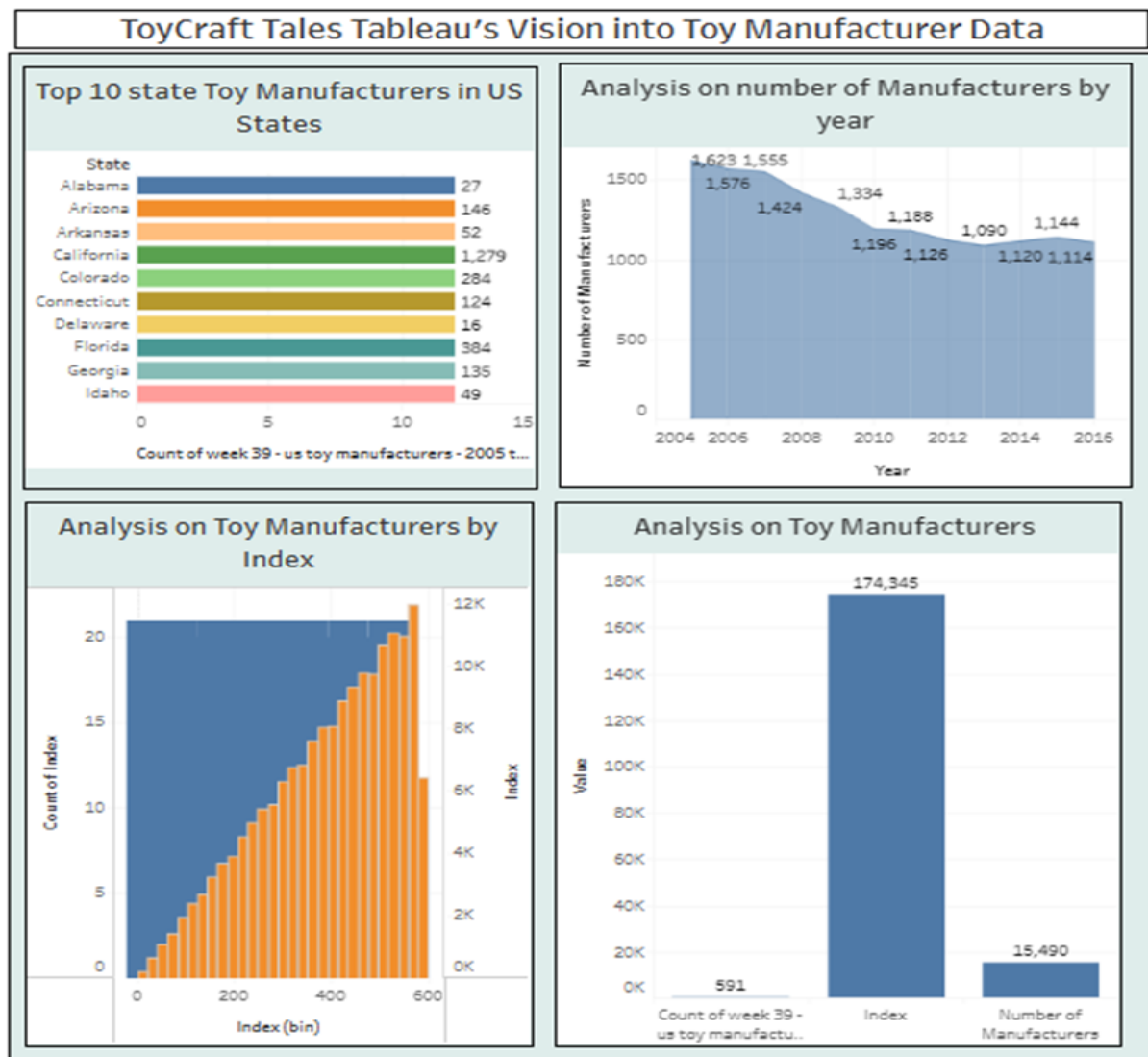
Table: week 39 - us toy manufacturers - 2005 to 2016

Columns:

index	int
State	text
Year	int
Number of Manufacturers	int



Utilization of Data Filters



No of Visualizations/ Graphs

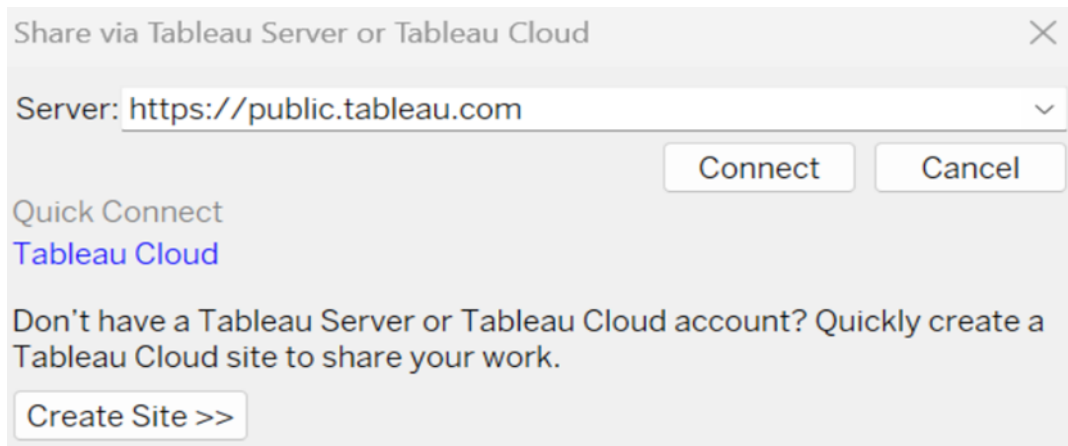
1. Analysis of the number of Manufacturers by Year
2. Analysis of Toy Manufacturer by Index
3. Analysis of toy Manufacturers in US state By Index
4. Analysis of Toy Manufacturers
5. Top 10 states toy manufacturer in US state

Web Integration

Publishing helps us to track and monitor key performance metrics, and to communicate results and progress. help a publisher stay informed, make better decisions, and communicate their performance to others.

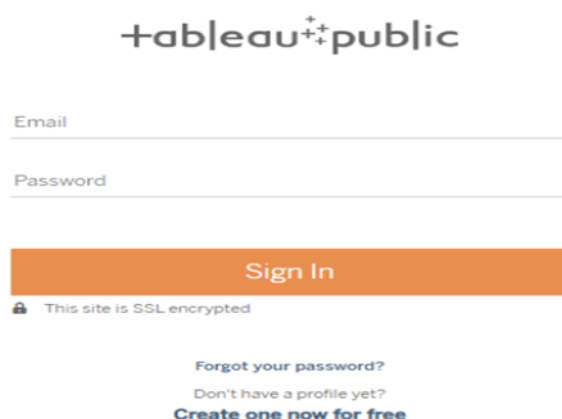
Publishing dashboard and reports to tableau public

Step 1: Go to Dashboard/story, click on the share button on the top ribbon

A screenshot of a web dialog box titled "Share via Tableau Server or Tableau Cloud". It features a close button (X) in the top right corner. Below the title is a "Server:" label followed by a text input field containing "https://public.tableau.com" and a dropdown arrow. To the right of the input field are two buttons: "Connect" and "Cancel". Below these is a "Quick Connect" section with a link "Tableau Cloud" in blue. At the bottom, there is a message: "Don't have a Tableau Server or Tableau Cloud account? Quickly create a Tableau Cloud site to share your work." followed by a "Create Site >>" button.

Give the server address of your Tableau public account and click on connect.

Step 2: Once you click on connect it will ask you for Tableau public user name and password

A screenshot of the Tableau Public login page. At the top is the "tableau public" logo. Below it are two input fields: "Email" and "Password". A large orange "Sign In" button is centered below the fields. At the bottom, there is a security notice: "This site is SSL encrypted". Below that are three links: "Forgot your password?", "Don't have a profile yet?", and "Create one now for free".

Once you log into your Tableau Public using the credentials, the particular visualization will be published into Tableau Public Note: While publishing the visualization to the public, the respective sheet will get published when you click on the share option.

Dashboard and Story embed with UI With Flask

 ToyCraft Tales

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ToyCraft Tales: Tableau's Vision into Toy Manufacturer Data

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 ToyCraft Tales

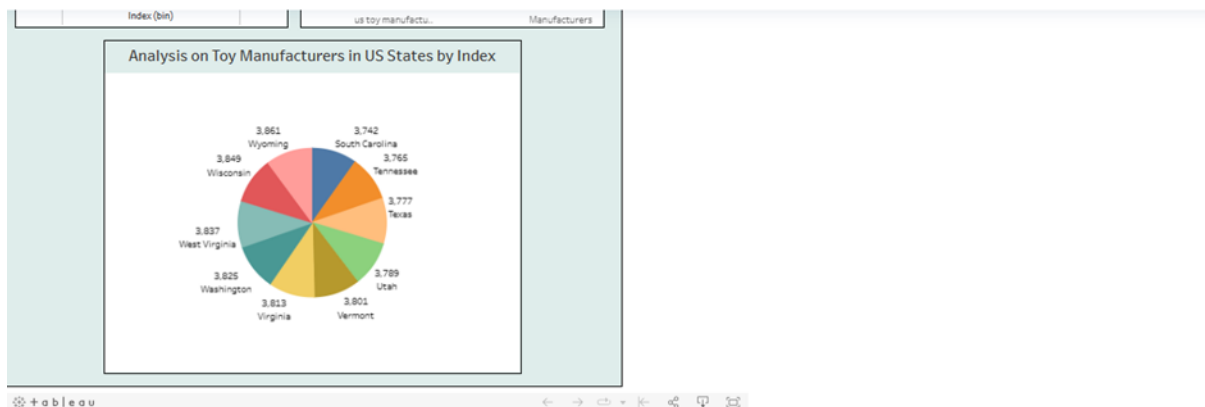
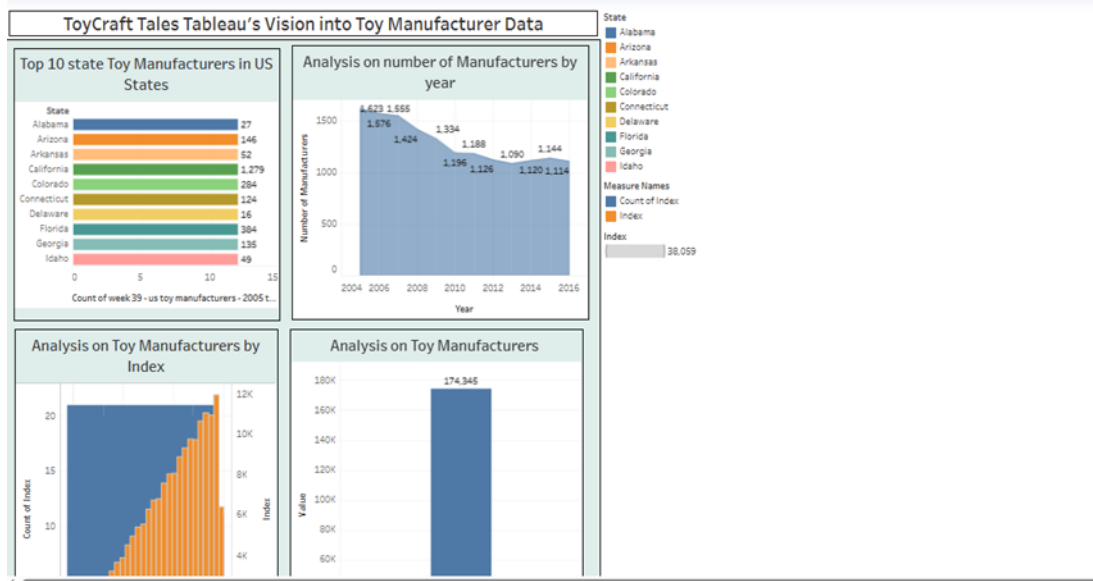
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WHO WE ARE

Toy manufacturing involves the process of designing, producing, and assembling toys for children and, in some cases, adults. The industry is diverse, encompassing a wide range of products, from traditional toys like dolls and action figures to modern electronic toys and games.

[Read More →](#)





ToyCraft Tales Tableau's Vision into Toy Manufacturer Data story

